

Part G : Feature Construction & Transformation

FunctionTransformer (Log Transform)

	annual_income	income_log
0	0.863137	0.622262
1	0.979218	0.682702
2	0.853882	0.617282
3	0.711704	0.537490
4	0.940591	0.662992

Reciprocal Transform

	annual_income	income_reciprocal
0	0.863137	0.536729
1	0.979218	0.505250
2	0.853882	0.539409
3	0.711704	0.584213
4	0.940591	0.515307

Square Root Transform

	annual_income	income_sqrt
0	0.863137	0.929052
1	0.979218	0.989555
2	0.853882	0.924057
3	0.711704	0.843626
4	0.940591	0.969840

Power Transformer

	annual_income	income_power
0	0.863137	0.169452
1	0.979218	1.244707
2	0.853882	0.097567
3	0.711704	-0.796784
4	0.940591	0.848875

column Transformer

```
ColumnTransformer  
(1000, 6)
```

New Features

Debt-to-Income Ratio

	loan_amount	annual_income	debt_to_income
0	0.504970	0.863137	0.585040
1	0.202809	0.979218	0.207114
2	0.520466	0.853882	0.609529
3	0.702479	0.711704	0.987038
4	0.339543	0.940591	0.360989

Average Monthly Transactions

	transaction_count	avg_monthly_transactions
0	95	15.833333
1	162	27.000000
2	11	1.833333
3	351	58.500000
4	398	66.333333

Spending-to-Income Ratio

	spending_ratio	annual_income	spending_to_income
0	54.536741	0.863137	63.184340
1	69.063641	0.979218	70.529365
2	36.848740	0.853882	43.154357
3	71.208527	0.711704	100.053539
4	44.621367	0.940591	47.439735